

ZAGROSIA

ENGINEERING

Website Design & Development

Internship Presentation — BCIT NMDW



zagrosia.ca — Insert live website screenshot

Full homepage, desktop view

Morsal Mozneb

Operations & Web Dev

Farnaz Gholami

UX/UI Designer

May – July 2026

Meet the Team



Morsal Mozneb

Operations & Front-End Developer

Next.js · TypeScript · Tailwind · Framer Motion



Farnaz Gholami

UX/UI Designer

Figma · Wireframing · Prototyping · Design Systems

The Company

Zagrosia Engineering Inc. / i3 Building Science & Consulting

35+

Years of
Expertise

3

Provincial
Licences

P.Eng

BC · AB
Sask

Founded by Seyed Hassan Mozneb, M.Sc., P.Eng., Zagrosia Engineering is a structural engineering firm serving developers, contractors, and building owners across Western Canada — combining 35+ years of expertise with AI automation and engineering education.



Company photo or project image

Insert a structural project photo

The Project

DURATION

9 weeks (May 10 – July 5, 2026)

HOURS

340+ combined hours

PAGES BUILT

8 full pages

DELIVERABLE

Live production website

8 Pages

01 Home

02 About

03 Services

04 Projects

05 AI Lab

06 Academy

07 Blog

08 Contact

Internship Experience

9 weeks · May – July 2026

Week 1 — Onboarding

Email Infrastructure

Microsoft 365 setup — info@zagrosia.ca, DNS via GoDaddy, team accounts configured.

LinkedIn Business Page

Created and launched the Zagrosia Engineering company page with branded profile and overview.

Competitor Research

Analyzed 5+ structural engineering websites in BC. Identified gaps and UX patterns to improve on.

Target Client Mapping

Defined ideal clients: project owners, developers, contractors. Informed navigation and content.



LinkedIn Business Page screenshot

Insert: zagrosia LinkedIn page



Microsoft 365 / Email setup screenshot

Insert: Outlook / DNS setup

UX Research & Planning

Farnaz — Week 1

Sitemap Planning

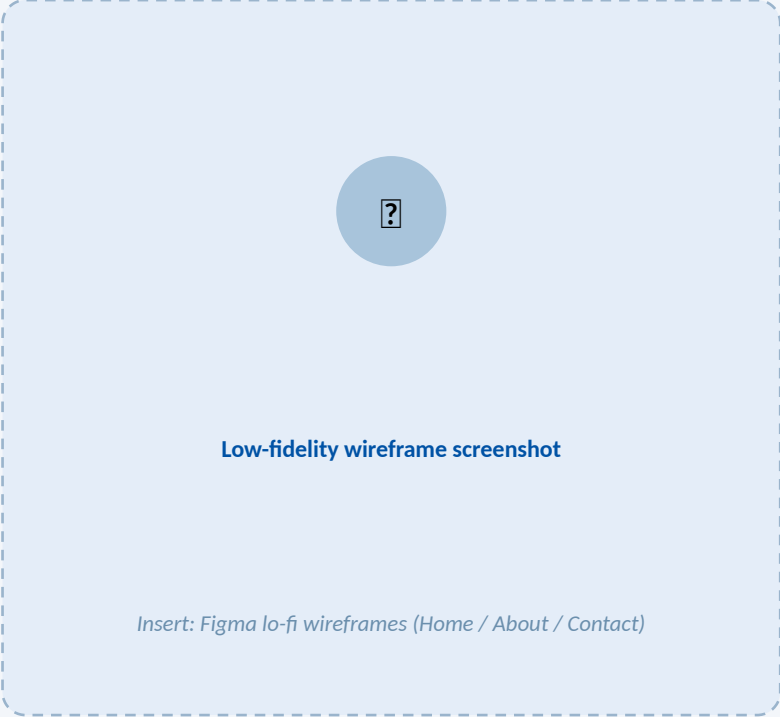
Defined site architecture: 8 primary pages with clear hierarchy and user journeys.

User Flow Mapping

Created detailed user flows for key conversion paths — from landing to contact form.

Low-Fi Wireframes

Completed lo-fi wireframes for Home, About, and Contact in Figma. Fast iteration to test layout logic.



Low-fidelity wireframe screenshot

Insert: Figma lo-fi wireframes (Home / About / Contact)

Weeks 2-3: Digital Operations

Morsal — content, brand, and SEO foundation

01

Content Strategy

Defined tone, messaging hierarchy, and key differentiators for a structural engineering audience.

02

Brand Documentation

Compiled brand guidelines with Farnaz's design system — colors, typography, logo, image standards.

03

SEO Foundation

Keyword research for BC and Alberta structural engineering. Metadata strategy for all 8 pages.

From Lo-Fi to Hi-Fi

Farnaz — Figma design process, weeks 2 through 4

STAGE 1

Low-Fidelity

Weeks 1-2

- Grayscale wireframes for all 8 pages
- Layout & information hierarchy validated
- 3-4 layout versions per key page



Low-Fidelity screenshot

Insert Figma stage 1 example

STAGE 2

Mid-Fidelity

Week 3

- Real content and spacing added
- Typography hierarchy locked
- Component patterns established



Mid-Fidelity screenshot

Insert Figma stage 2 example

STAGE 3

High-Fidelity

Weeks 4-5

- Full color, imagery, micro-copy
- Interactive Figma prototype
- Client review-ready deliverable



High-Fidelity screenshot

Insert Figma stage 3 example

Client Review Meeting

Week 5 — Presenting the Hi-Fi prototype to Homayoun Mozneb, P.Eng.



Figma Hi-Fi prototype screenshot

Insert: interactive prototype — full 8-page Figma walkthrough

Key Feedback

- Refine hero typography
- Add P.Eng credential badges
- Strengthen AI Lab section
- Expand services detail copy

• Mobile responsiveness

Takeaway: Real client feedback revealed priorities that no spec document could. The session taught us to design for communication, not just aesthetics.

Development Begins

Morsal — Week 4: All 8 pages built in a single intensive sprint

AI Dev Stack

Codex

AI code generation — boilerplate and scaffolding

Claude Code

Complex logic, refactoring, multi-file coordination

V0 by Vercel

UI component generation from design descriptions

GitHub Copilot

Inline completions while writing TypeScript



Code editor + website preview

Insert: VS Code split with zagrosia.ca preview

Pages Completed:

Home ✓

About ✓

Services ✓

Projects ✓

AI Lab ✓

Academy ✓

Blog ✓

Contact ✓

Refinement & Polish

Desktop layouts, animations, QA, and launch

Wk 6

Desktop
Layouts

Desktop-First Overhaul

Wider containers, multi-column grids, larger typography. Rebuilt every page for 1280px+ screens.

Framer Motion Animations

Scroll-triggered entrance animations — fade-up, blur-in, scale. Stagger effects across every section.

Magnetic CTA Buttons

Custom component: magnetic pull + shimmer on hover + spring press on click. Three layered effects.

Wk 7

Card Design
& Shadows

Wk 8

Scroll
Animations

Wk 9

QA & Launch

?

Website animations / mobile view

Insert: before-after desktop vs. mobile screenshot

Showcase Your Work

zagrosia.ca — 8 pages, live in production

zagrosia.ca — At a Glance

8

Pages

35+

Components

01 Home

02 About

03 Services

04 Projects

05 AI Lab

06 Academy

07 Blog

08 Contact

?

zagrosia.ca — Full homepage screenshot

Insert: desktop view of homepage — hero to footer

Home Page

First impression — hero, stats, platform sections

Hero Section

Animated headline, "Engineered to Solve" tagline, two magnetic CTA buttons. Background cycles through engineering imagery.

Engineering Stats

35+ years, P.Eng credentials — count-up on scroll. Social proof built into the visual rhythm.

AI Lab Preview

Dark-themed section: 500+ videos, 150+ subjects in the AI education library.

EGBC Exam Prep

Teaser for the Academy — exam prep for engineering licensure candidates in BC.



Home page — Hero section screenshot

Insert: desktop hero section at full width

Services Page

Six structural engineering services

01

Structural Design

Residential & commercial systems

02

Seismic Assessment

Vulnerability studies + retrofit

03

Rehabilitation

Assessment + repair specs

04

Racking Systems

Industrial racking + mezzanine

05

Building Envelope

Enclosure + moisture assessment

06

Specialty Reviews

Peer review + letter of assurance



Services page screenshot

Insert: services page — desktop full view

Projects Portfolio

Filterable card layout — real completed engineering projects

Card Layout

Project type, location, scope, and featured image per card.

Filter System

Filter by Residential, Commercial, Industrial, or Seismic.

Hover States

Cards reveal detail overlay and CTA on hover.

Project Hero

Animated stat callouts and category breakdown at top.



Projects page screenshot

Insert: project cards grid with filter tabs

AI & Automation Lab

Where structural engineering meets AI

500+

Videos

150+

Subjects

200+

Views

Platform Capabilities

- Custom Python scripts for structural calculation automation
- VBA macros for Excel-based engineering workflows
- AI educational platform for engineering topics



AI Lab page screenshot

Insert: AI Lab page — full desktop view with stats

Academy — EGBC Exam Prep

Structured learning platform for engineering licensure

What It Offers

- Structured roadmap for EGBC PPE exam
- Video lessons for exam-critical concepts
- Practice questions + self-assessment
- Downloadable resources + formula sheets
- Progress tracking and milestones



Academy / EGBC Exam Prep screenshot

Insert: Academy page — course cards and study roadmap

Blog, About & Contact

Blog



Blog page screenshot

Insert: Blog page view

About



About page screenshot

Insert: About page view

Contact



Contact page screenshot

Insert: Contact page view

Technology Stack

Every tool chosen for a reason

FRAMEWORK

Next.js 15

App Router, SSR, file-based routing

TypeScript

Type safety across all components

STYLING

Tailwind CSS

Utility-first, mobile-first

Framer Motion

Physics-based scroll animations

DESIGN

Figma

Full design system + prototypes

React Icons

Consistent iconography

DEVOPS

GitHub

Version control, CI/CD trigger

Netlify

Auto-deploy on push to main

AI TOOLS

Codex / Claude Code

AI-assisted development

V0 by Vercel

UI component generation

Signature Animations

The interactions that make zagrosia.ca feel alive

Scroll Reveals

Framer Motion · whileInView

Fade-up, blur-in, scale — staggered delays cascade down each section.

Magnetic CTA Buttons

useMotionValue · useSpring

Spring physics drift + shimmer sweep + 0.93 scale press. Layered effects.

Count-Up Numbers

useInView · rAF

Stats count from zero on scroll-enter. Triggers once, reinforces credibility.

Reduced Motion

useReducedMotion()

All animations silently disable if the OS accessibility setting is active.



Animation in action

Insert: GIF or screenshot of magnetic button + scroll reveal on homepage

Design System

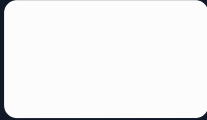
Farnaz — the visual foundation

COLOR PALETTE



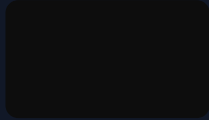
#0052A5

Navy · Primary



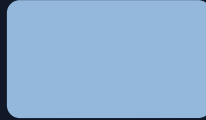
#FCFCFC

White · BG



#0D0D0D

Black · Dark



#94B8DC

Blue · Accent

TYPE SCALE

Display / H1

40–52px

Section / H2

24–32px

Body

12–16px

Caption

8–11px



Figma design system screenshot

Insert: Figma file — color tokens, type styles, component library

What We Learned

Technical skills, collaboration, and real-world growth

Technical Growth — Morsal

Next.js 15 App Router

Built a production app with App Router — server/client components, route groups, metadata API.
Not tutorials: real architecture.

TypeScript in Practice

Went from avoiding TS to relying on it. Interfaces and type guards caught real bugs before production.

AI as a Development Tool

Learned to use Codex, Claude Code, and V0 as precision instruments. The skill is knowing what to prompt and when to discard the output.

Git & CI/CD Pipeline

Managed the full GitHub → Netlify pipeline. Diagnosed real deployment failures. Production DevOps, not sandbox practice.

Framer Motion Animations

Spring physics, scroll triggers, gesture animations. Custom CtaButton with three layered motion effects + accessibility support.

Design Growth — Farnaz

Research-Driven Design

Validated every decision with research — competitor analysis, user flows, sitemap — before opening Figma. Design without research is guesswork.

Figma Prototyping

Advanced techniques: interactive components, smart animate transitions, reusable tokens. Client clicked through the prototype like a real site.

Client Communication

Presenting to a real client differs from a school critique. Frame decisions in business terms. Listen for the actual need behind the stated feedback.

Responsive Design

Mobile first, then desktop. Maintaining hierarchy across breakpoints requires a systematic approach to spacing and type scale.

Design-to-Dev Handoff

Tighter collaboration with Morsal produced better outcomes than any clean handoff would have.

Designer + Developer Collaboration

What we learned from building together



What surprised us:

Design decisions obvious in Figma are often hardest to implement — and implementing them reveals better alternatives.

Biggest challenge:

Keeping design and code in sync across 9 weeks without a formal handoff tool. We solved it with frequent syncs and shared Figma access.

What we'd do differently:

Establish a component naming convention from day one. Naming mismatches between Figma and React caused friction mid-project.

Key Insights

01

[Morsal] AI tools amplify judgment — they don't replace it

The bottleneck shifted from 'can I write this?' to 'is this the right thing to write?'

02

[Farnaz] Real clients operate differently from academic briefs

The brief changed. Priorities shifted. A 2-hour review reshaped the product more than weeks of planning.

03

[Morsal] Polish is where the professional gap lives

The gap between a working site and a professional product is animation timing, spacing, and accessibility.

04

[Farnaz] Design and code are two halves of one discipline

The best decisions happened when we were in the same conversation. Silos produce friction.

Looking Ahead

From internship to career

It's Live.

zagrosia.ca

Real Client

A practicing P.Eng now has a digital presence that reflects the full scale and quality of their firm.

Real Stack

Next.js 15 · TypeScript · Tailwind CSS · Framer Motion · GitHub → Netlify CI/CD. Production infrastructure.

Real Handoff

Delivered with documentation, admin access, and a pipeline that auto-updates on every git push.



zagrosia.ca — live website screenshot

Insert: full desktop homepage in browser — Chrome address bar visible showing zagrosia.ca

What's Next

Morsal

- Continue zagrosia.ca — CMS integration, blog automation, advanced AI features
- Front-end dev roles in BC tech — Next.js, TypeScript, animation-forward products
- Apply AI-assisted workflow to freelance and agency projects

Farnaz

- Build the Zagrosia design system as a reusable Figma library for firm materials
- UX/UI design roles at product companies and agencies in Metro Vancouver
- Continue developing interaction design and design-to-dev handoff expertise

Thank You

Questions & Discussion

Morsal Mozneb · Farnaz Gholami

BCIT New Media Design & Web Development · 2026

zagrosia.ca